17 August 2026

Dr. Nur Zincir-Heywood
Associate Editor-in-Chief
IEEE Transactions on Network and Service Management

Dear Dr. Zincir-Heywood,

Thank you for the decision on TNSM-2026-11591, "Learning to Route Under Load: An Inductive Graph Network that Amortizes Congestion-Aware Traffic Engineering for LEO Mega-Constellations," and for the invitation to submit a new manuscript. We are grateful to Associate Editor Dr. Tianqi Yu and to the three reviewers, whose comments were specific enough to be acted on rather than merely acknowledged.

We should say at the outset that this is not the same paper with corrections. Two of the reviews asked questions we had not asked ourselves, and answering them changed our conclusions. The title has changed with them, to "A Reality Check on Learned Traffic Engineering for LEO Mega-Constellations." The previous one promised a method; the paper now reports a measurement, and the title should say which of the two a reader is being handed.

Reviewer 3 pointed out that our headline gap is produced by a road-traffic congestion function evaluated where a packet link cannot operate: the excess offered to a link is lost, not delayed. Enforcing capacity feasibility reduces that gap from 80.5% measured as travel time to 29.8% measured as delivered rate. Reviewer 2 pointed out that our comparators were weak, and asked for a strong baseline at a matched computational budget. Supplying one, a congestion-blind multipath split costing a single pass, showed that our method's advantage is decisive on the constellation it was trained on and level with that baseline on constellations it was not.

Those two results do not support the four contributions the original manuscript claimed. Rather than restate weakened versions of them, we have rewritten the paper as what the measurements actually support: a study of what learned congestion prices buy in LEO routing, where a blind baseline is already sufficient, and why the boundary falls where it does. The system model, the simulator and the learned method are unchanged. The claims built on them are not, and the resubmitted manuscript reports smaller numbers than the original in three places.

We also owe the Editor a correction. Following the reviewers' and your own remarks on related work, we found arXiv:2601.21921, which trains a graph network to infer per-link congestion prices from the constellation state in a single forward pass for LEO mega-constellations. That is the same core idea as our original contribution, it was public five months before our submission, and we had not cited it. It is now cited and discussed, and it is part of why the method framing is no longer the right one for this work.

Marked-up copy. We enclose both a clean manuscript and a marked-up copy. In the marked-up copy, blue text is new in this manuscript and red text was carried over from the previous submission and removed; unmarked text is unchanged. We should be straightforward about one limitation of that copy. This is a rewrite rather than an edit: of the previous manuscript's body sentences, 42% are gone and 56% of the present one is new, and at that distance an automatic comparison cannot align the two documents. The blue marking is reliable, but most deleted sentences could not be placed in context and so do not appear in red. Rather than leave that gap implicit, we say so at the head of the marked-up copy, and the response letter lists what was removed and why. We have used colour alone rather than underlining or strikethrough so that the marked pages read at the same size and weight as the final paper.

Data and code. The simulator, the learned model, every experiment including those added in response to the reviewers, and the scripts that generate each reported number are available at `https://github.com/haodpsut/qwgnn-leo-routing`. Every headline number in the paper is tied to the file, column and aggregation that produce it by a claims manifest, and a script recomputes all of them.

We hope the reviewers find the changes responsive. Where we have declined a suggestion, in particular four suggested citations from Reviewer 1 that concern RIS and anti-jamming rather than routing, we say so and explain why rather than adding them silently.

Sincerely,

Phuc Hao Do, PhD